

# Input, Process, Output

Project: Helpful Classroom Sorter

## 1. Your mission

### Your learning target

I can trace information through a system without confusing information with a physical action.

I can explain which details a representation preserves or discards.

I can locate the earliest stage at which an incorrect result could have arisen.

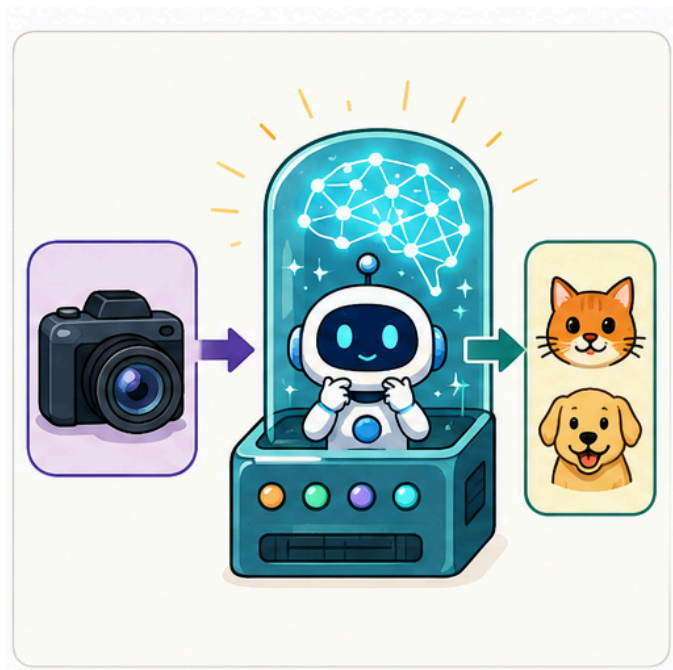
### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

A system transforms information; it does not simply 'know.'



LEARNING TARGET A library kiosk receives 'mystery', filters a catalogue by genre and displays three titles. No speaker is attached. Draw its information path. A visitor says it read three books aloud. Which stage contradicts that claim?

Learner name:

### Readiness for the connected explanation

In last week's bus case, the forecast number and the display colour came from different decisions. Which decision produced each output?

### This week's end goal

By the end, I can trace an information path and name what a summary loses.

## 2. Notice the evidence

PICTURE STORY

### Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



MYSTERY A music helper receives listening choices, compares them with a ranking process, and suggests a song. The suggestion appears only after information has moved through the system.

### What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

GOOD THINKING A prediction can change when stronger evidence appears.

## A closer explanation

A library kiosk asks Sam for a genre. Sam selects mystery. The kiosk compares that preference with catalogue tags and displays three book titles. Sam says the kiosk has read three books aloud. That conclusion adds an action that the description never supplied: displaying a title is different from producing spoken audio.

An information path names what enters each stage, what happens there and what leaves it. Treat input and output as roles relative to a stage. A list of titles is the output of the search, but it could become the input to a separate text-to-speech stage. A component can therefore receive the output of another component without doing the same job.

The kiosk's genre-matching operation can be implemented with a written rule. Nothing in this description establishes that a model was trained. This connects to Week 1: tracing information tells us what a process needs to do, while mechanism evidence tells us how it does it.

## 3. See how it works

VISUAL MECHANISM

**Input → Process → Output**

Follow the information instead of imagining magic inside the machine.



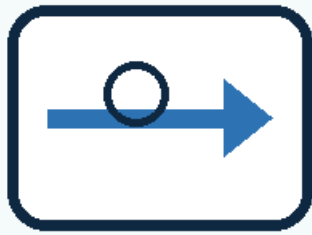
## INPUT

songs played, liked, or skipped



## PROCESS

compare clues and rank possible songs



## OUTPUT

a suggested song or a not-sure result

INPUT songs played, liked, or skipped

PROCESS compare clues and rank possible songs

OUTPUT a suggested song or a not-sure result

CORE IDEA A system transforms information; it does not simply 'know.'

UNDER THE HOOD Input, process, and output are functional roles. The process may contain human choices, fixed rules, calculations, retrieval, or a learned model. Naming the three stages prevents magical explanations of computer behavior.

Draw a fresh example with all three stages:

### A closer explanation

#### Two possible library information paths

##### Input · Process · Output

**Input:** Genre request mystery and catalogue tags

**Process:** Keep matching catalogue records

**Output:** Three matching titles

---

**Input:** Three titles from the first stage

**Process:** A separate speech system if available

**Output:** Spoken titles rather than whole books

---

**Input:** A selected title and an available copy

**Process:** A person completes the borrowing process

**Output:** A recorded loan and a borrowed copy

---

The first row is the kiosk as described. The later rows require additional components and actions. Arrows show information dependencies; they do not establish that every stage has actually run.

Notice the difference between a title, a description and an entire book. All three can be text, but they represent different amounts of information. Even a speech stage receiving three titles can speak only those supplied titles; it has not thereby read the full books. Always identify the content of an input, not just its data type.

A recommendation is also different from a loan. Selecting a title does not reserve a copy unless an authorised reservation step exists. Keeping these distinctions clear prevents us from crediting a system with actions it did not take.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

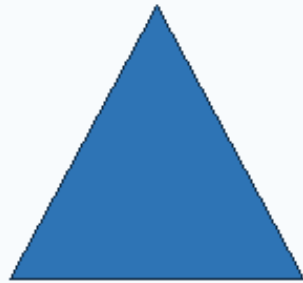
### PICTURE GALLERY

#### Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.



**Camera app**



**Music ranking**



**Song suggestion**



**No listening history**

CASE 1 — Camera app Clue: photo enters before any result. Meaning: input.

CASE 2 — Music ranking Clue: choices are compared and scored. Meaning: process.

CASE 3 — Song suggestion Clue: a result is shown to the user. Meaning: output.

CASE 4 — No listening history Clue: important input is missing. Meaning: not-sure case.

BOUNDARY The output is not proof that the system understood the input as a person would.

Which two pictures are easiest to confuse? What evidence separates them?

## Change one thing in the pictured case

Change: Store red count alongside total.

Prediction: The records become (6,2) and (6,5).

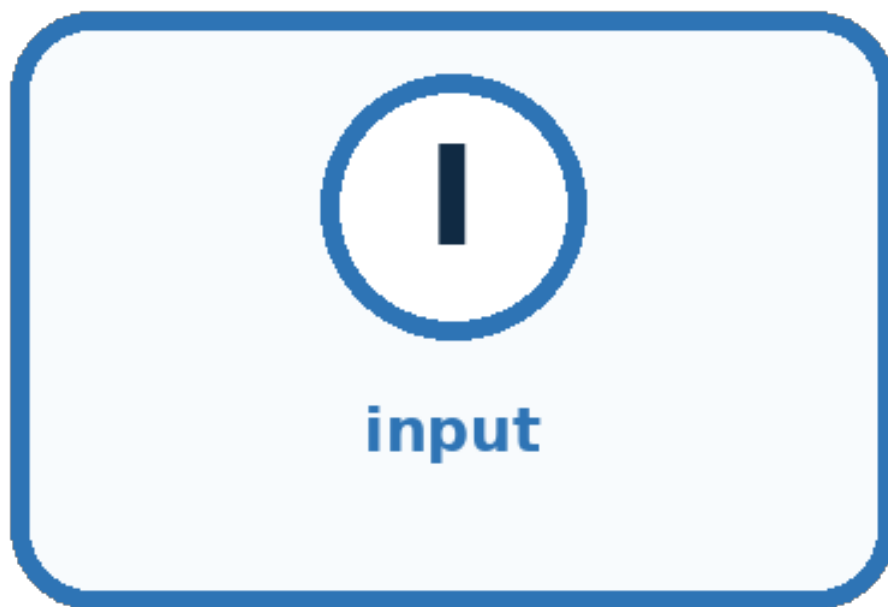
Reason: Blue count can now be recovered by total minus red, assuming these are the only colours.

## 5. Words that unlock the idea

### VOCABULARY

#### Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





**process**



**output**



### Word · Meaning · My example

**Word:** input

**Meaning:** information entering a system

**My example:** \_\_\_\_\_  
\_\_\_\_\_

**Word:** process

**Meaning:** the steps that transform information

**My example:** \_\_\_\_\_  
\_\_\_\_\_

**Word:** output

**Meaning:** the result produced by the system

**My example:** \_\_\_\_\_  
\_\_\_\_\_

**Word:** trace

**Meaning:** follow information through each stage

**My example:** \_\_\_\_\_  
\_\_\_\_\_

### Misconception detective

CLAIM TO REPAIR The computer already knows the answer before it receives input.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

## Words used in the closer explanation

### Word · Meaning

**Word:** Representation

**Meaning:** A way of recording information so that it can be stored, communicated or processed

---

**Word:** Information loss

**Meaning:** Details that cannot be recovered from the retained representation

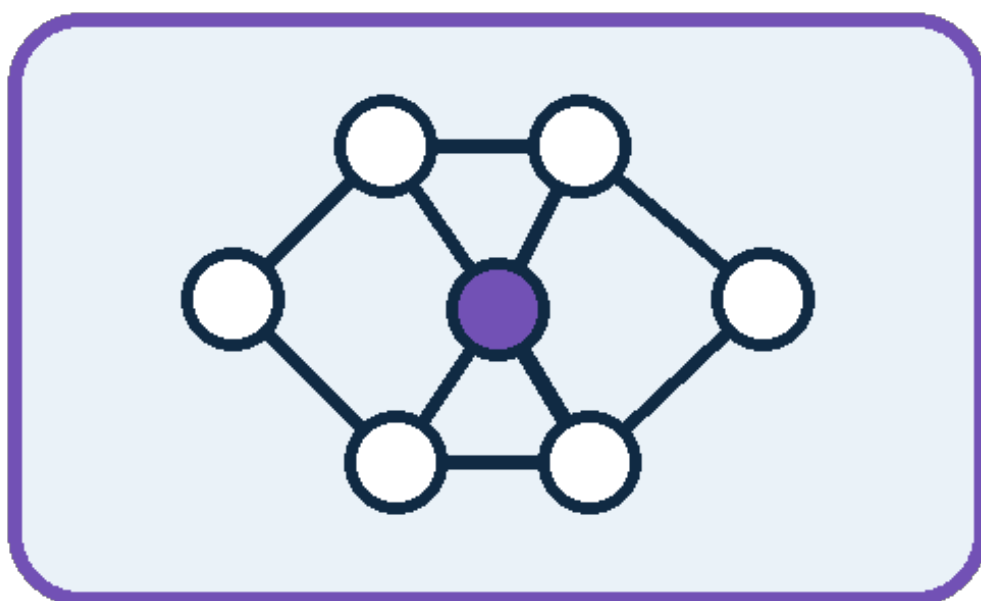
---

## 6. Follow a worked example

WORKED EXAMPLE

### Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE A music helper receives listening choices, compares them with a ranking process, and suggests a song. The suggestion appears only after information has moved through the system.

STEP 1 — OBSERVE songs played, liked, or skipped

STEP 2 — TRACE compare clues and rank possible songs

STEP 3 — CONCLUDE a suggested song or a not-sure result

MODEL REASONING Start with what enters, name the actual transformation rather than saying the system thinks, and identify the result. If the process is unknown, state what evidence is missing instead of inventing a mechanism.

DO NOT OVERCLAIM The output is not proof that the system understood the input as a person would.

What new evidence could change this answer?

## A closer explanation

Consider two catalogue records. Book A has 90 pages and is about a child solving a seaside mystery. Book B has 90 pages and is about a lost spaceship. If we keep only page count, both become the number 90. The number is useful for a length limit, but it cannot tell us which story is a mystery. Information that was never retained cannot be recovered just by calculating harder.

**Representation kept · Question it can help answer · Question it cannot settle alone**

**Representation kept:** Page count 90

**Question it can help answer:** Is the book under 100 pages?

**Question it cannot settle alone:** Is its story a mystery?

---

**Representation kept:** Genre tag mystery

**Question it can help answer:** Does it match a mystery request?

**Question it cannot settle alone:** Will this particular reader enjoy it?

---

**Representation kept:** Short summary

**Question it can help answer:** What does the description say happens?

**Question it cannot settle alone:** What exact sentence occurs on page 42?

The phrase cannot settle alone matters. A genre tag might be inaccurate, and a reader might enjoy an unfamiliar genre. A representation is useful only in relation to a question and its limits. More detail is not automatically better: unnecessary personal information can add risk without helping the task.

**Learn from the worked case**

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

**A pictorial worked case: Input, Process, Output**

A fictional classroom cupboard records coloured pencils, then reports just a total.

**The facts we will use**

Box A has 2 red and 4 blue pencils. Box B has 5 red and 1 blue.

The process adds colour counts and stores only the total, not the two counts.

AI ACADEMY / AI-1/2

**Different colour counts, same total**

|       | Red | Blue | Total |
|-------|-----|------|-------|
| Box A | 2   | 4    | 6     |
| Box B | 5   | 1    | 6     |

Read a cell using both its row and column labels.

## Input, Process, Output

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Read colours

A: red 2, blue 4. B: red 5, blue 1.



#### 2. Add counts

A:  $2+4=6$ . B:  $5+1=6$ .



#### 3. Store total

Both records become total 6.



#### 4. Ask a new question

The saved totals cannot tell which box has more red pencils.

### CHANGE ONE THING

Store red count alongside total.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Read colours

**What happens and why:** A: red 2, blue 4. B: red 5, blue 1.

---

**Step:** 2. Add counts

**What happens and why:** A:  $2+4=6$ . B:  $5+1=6$ .

---

**Step:** 3. Store total

**What happens and why:** Both records become total 6.

---

**Step:** 4. Ask a new question

**What happens and why:** The saved totals cannot tell which box has more red pencils.

---

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct inputs lead to the same stored output. Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally.

### What this example does not establish

The two-colour assumption matters. Extra colours would prevent inferring blue from total and red alone.

## 7. Find and repair a mistake

COMPARE

### Same Result, Different Reason

A comparison helps reveal the true concept boundary.

#### Case · Evidence · What it shows

**Case:** Camera app

**Evidence:** photo enters before any result

**What it shows:** input

---

**Case:** Music ranking

**Evidence:** choices are compared and scored

**What it shows:** process

---

**Case:** Song suggestion

**Evidence:** a result is shown to the user

**What it shows:** output

---

**Case:** No listening history

**Evidence:** important input is missing

**What it shows:** not-sure case

---

## Repair two tempting claims

CLAIM A The computer already knows the answer before it receives input.

Repair:

CLAIM B The screen is the process because that is where the result appears.

Repair:

STRONG ANSWER FRAME I claim \_\_\_\_\_ because the evidence shows \_\_\_\_\_. The process is \_\_\_\_\_. One limit is \_\_\_\_\_.

## A closer explanation

The kiosk is asked for mystery books under 100 pages but displays a 140-page mystery. Before changing everything, inspect the intermediate records. Was the maximum entered as 100? Did the catalogue say 140? Did filtering preserve the wrong record? Did the display show a title different from the filtered result? The first mismatch between the intended and actual path helps locate the problem.

A correct-looking final screen can hide an earlier error. A calculator can correctly multiply a number that was transcribed incorrectly, and a warning rule can correctly use an incorrect forecast. Write down intermediate values so you can test each connection.

**Explain why a system that displays a book title has not necessarily spoken, reserved or read the book**

---

---

---

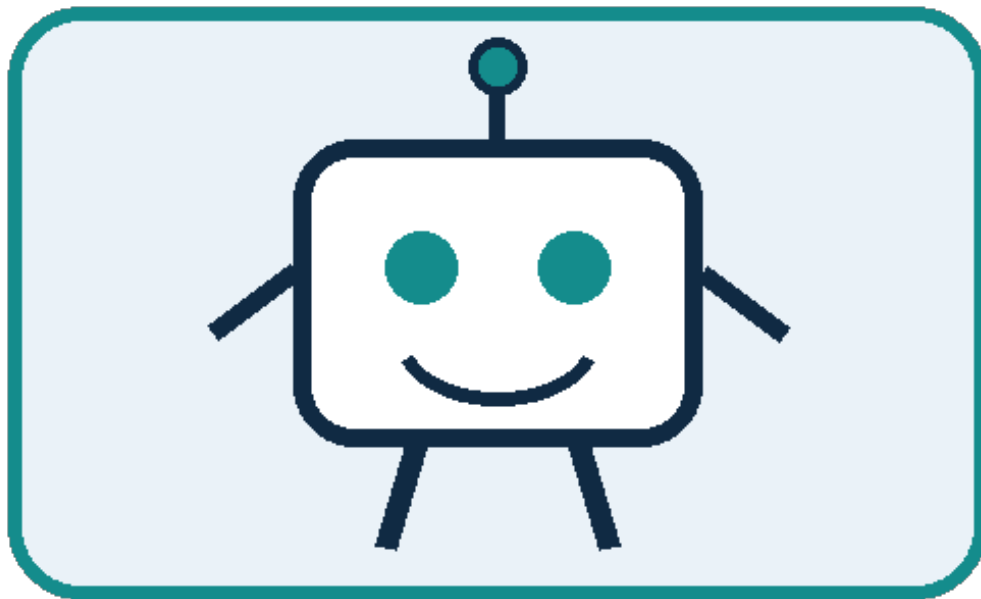
---

## 8. Practise together

GUIDED PRACTICE

## Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A system transforms information; it does not simply 'know.'

Use a fresh example to explain input.

Trace a new input → process → output example.

Find and repair the error in: The screen is the process because that is where the result appears.

Invent one edge case and name the safe response.

**PARTNER CHECK** Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

### A closer explanation

A greenhouse records morning temperatures of 18, 22 and 26 degrees C, then retains only the largest value, 26. Can the stored value prove that the temperature rose steadily? No. The sequence 22, 18, 26 has the same largest value but includes a decrease. These two sequences form a counterexample to the claim that the largest value determines the order.

To solve this, first state exactly what was kept: one maximum. Then ask what the claim requires: the order of several readings. Finally construct two possible originals that share the retained value but give different answers to the claim. If both originals fit the record, the record alone cannot decide between them.

**A record keeps only a visitor count of 3. Give two different arrival schedules consistent with it**

**and name one fact the count cannot reveal**

---

---

---

---

---

### Practise with less help: Finish the visitor example

Use workbook W2 for the connected example “Representations and information paths”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

#### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

---

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Representations and information paths. Proposed explanation: “Every output is the final action”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

#### My explanation using a pictured step and the given facts

---

---

---

---

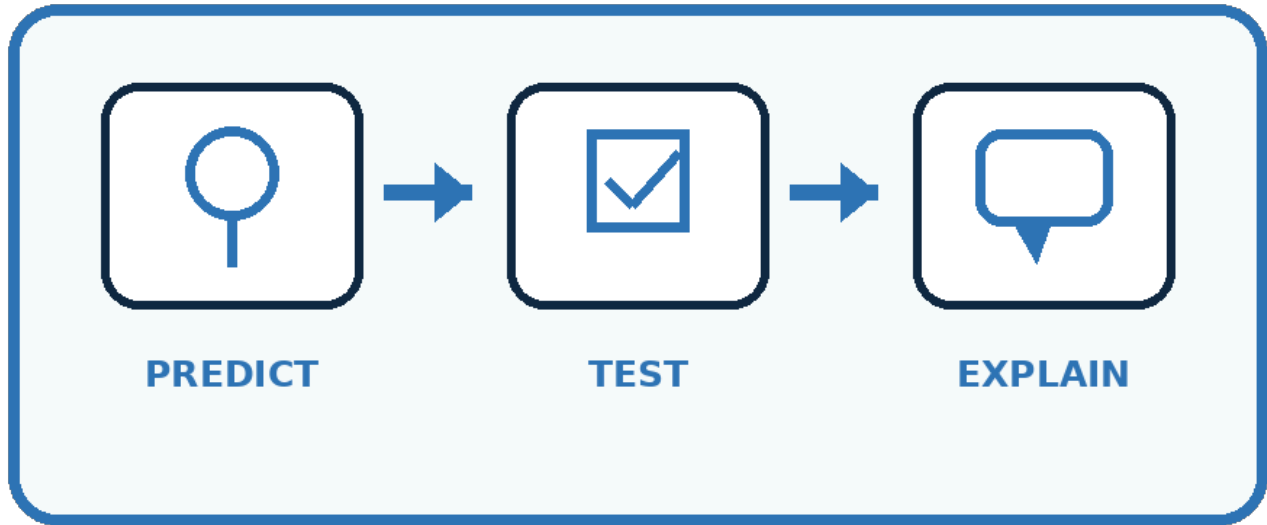
This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

EXPLORER LAB

### Human Music Recommender

Predict first, test honestly, and explain what the evidence means.



State your prediction before the result is known.

Complete the task: Trace a music recommender and draw all three stages.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

**Case · Prediction · Observed result · What it means**

**Case:** Normal

**Prediction:** \_\_\_\_\_

**Observed result:** \_\_\_\_\_

**What it means:** \_\_\_\_\_

\_\_\_\_\_

**Case:** Edge

**Prediction:** \_\_\_\_\_

**Observed result:** \_\_\_\_\_

**What it means:** \_\_\_\_\_

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

### Plan → try → check

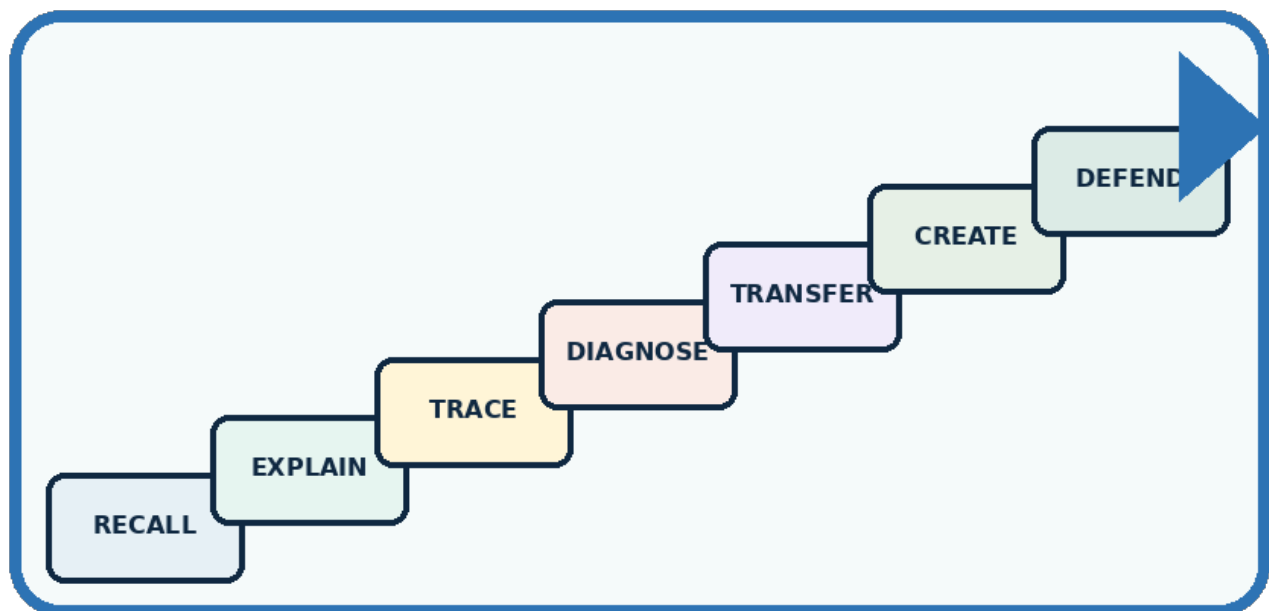
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

### PRACTICE LADDER

#### Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define input.

EXPLAIN A system transforms information; it does not simply 'know.'

TRACE Show the input, process, and output.

DIAGNOSE Repair: The computer already knows the answer before it receives input.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

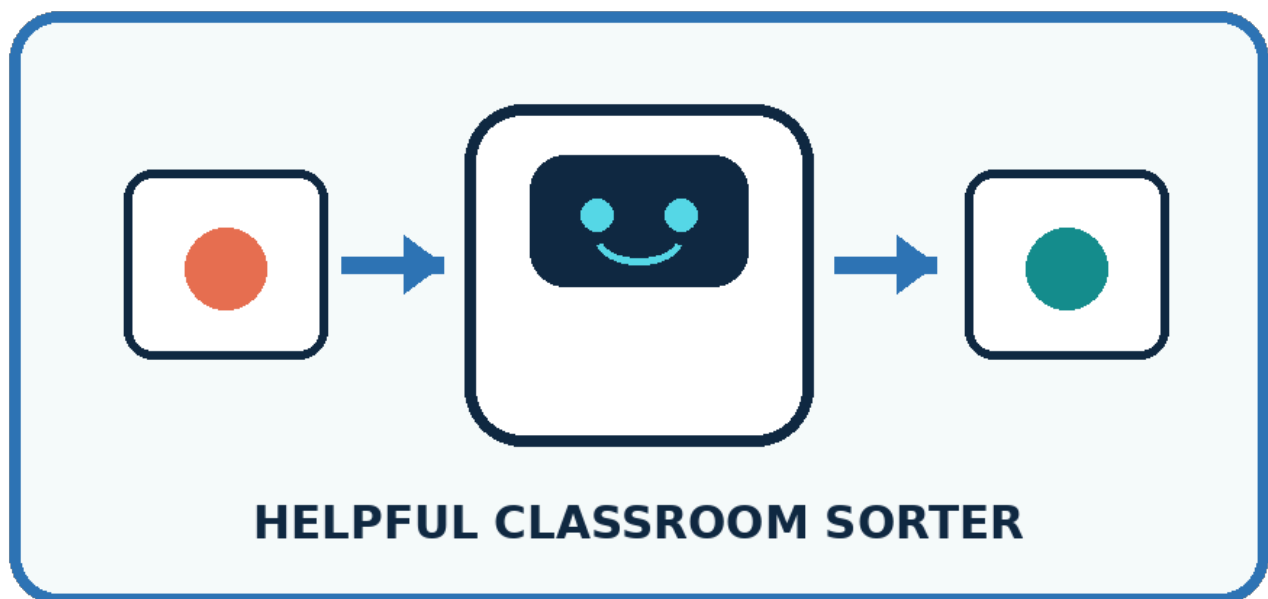
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

## 11. Connect your project

PROJECT

### Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



THIS WEEK'S CHECKPOINT Draw the Classroom Sorter's first input-process-output map and show where a person may review the result.

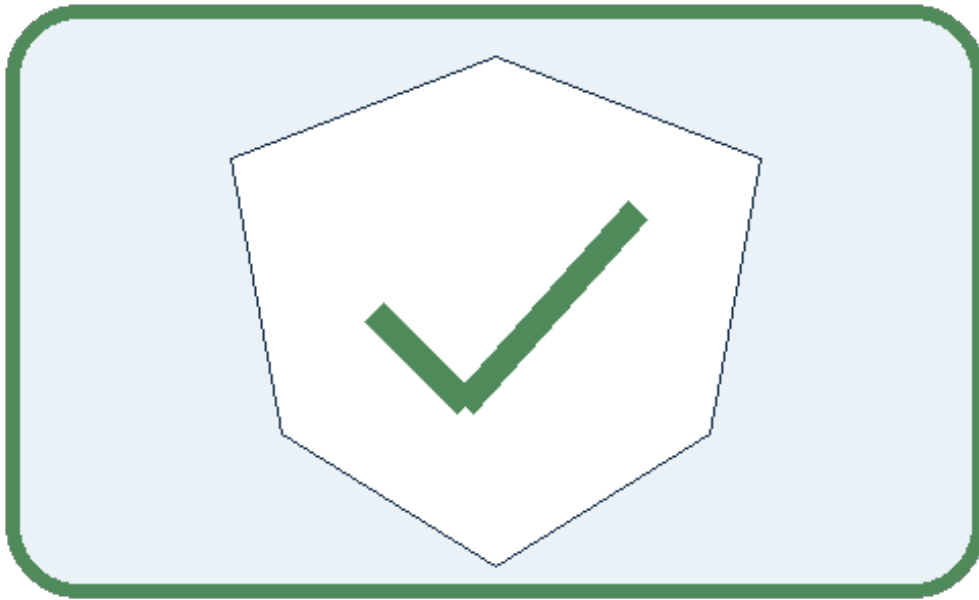
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



**RESPONSIBLE DESIGN** The output is not proof that the system understood the input as a person would.

My project decision:

### **Evidence for your project**

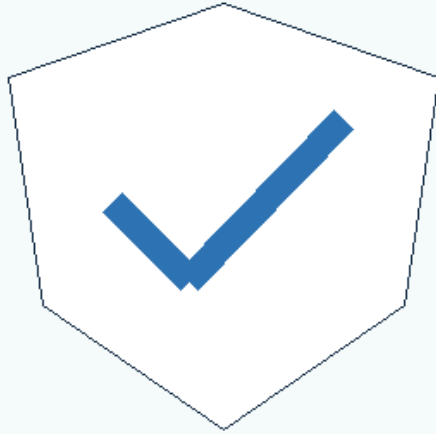
Choose sorter inputs that retain the clues needed for its decision; draw the transformation and identify discarded details.

## **12. Show what you know**

MASTERY

### **Fresh-Case Independent Proof**

Complete this after teaching and guided practice have ended.



## INDEPENDENT PROOF

**REQUIRED PROOF** Trace a fresh system with an accurate input, process, and output, then explain one missing-input case.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

**SELF-CHECK** Start with what enters, name the actual transformation rather than saying the system thinks, and identify the result. If the process is unknown, state what evidence is missing instead of inventing a mechanism.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

**NEXT BRIDGE** Carry today's evidence habit into: Sensors Are Messengers.

## 13. Study a complete case

### Week 2 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This

case extends the illustrated lesson.

**The situation**

A playlist helper receives a mood, matches song tags and displays three titles.

**The question**

Name input, process and output. Is displaying titles the same as playing audio?

**Worked explanation**

Input is mood; process is tag matching; output is titles. Audio playback is another step. Matching can use fixed rules; the description alone does not establish machine learning.

**Explain it in your own words**

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

---

---

---

---

---

**Trace the supplied case visually**

**From mood to song titles**

**Evidence or stage · Value or result**

**Evidence or stage:** Mood

**Value or result:** Input from listener

---

**Evidence or stage:** Tag matching

**Value or result:** Process chooses matches

---

**Evidence or stage:** Three titles

**Value or result:** Displayed output

---

**Evidence or stage:** Audio playback

**Value or result:** A separate possible step

---

A playlist helper receives a mood, matches song tags and displays three titles.

Name input, process and output. Is displaying titles the same as playing audio?

Input is mood; process is tag matching; output is titles. Audio playback is another step. Matching can use fixed rules; the description alone does not establish machine learning.

## 14. Try a new case independently

### Independent case: Investigate a new loss of information

A classroom device receives four scores and stores only their total, 20. A student claims that every score must be 5. Give two different lists of four nonnegative whole-number scores with total 20. Explain precisely what the stored total establishes and what it does not.

#### My independent explanation, trace and limits

---

---

---

---

---

---

---

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 2 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A classroom fan receives a temperature reading, compares it with 25 degrees C, and turns on. Name the input, process and output. What should happen if the reading is missing?

My answer, with a trace, calculation or supporting evidence

**Write or draw your reasoning and evidence.**

---

---

---

---

---

### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

---

---

---

---

---

### Connect to your project

Draw the Classroom Sorter's first input-process-output map and show where a person may review the result.

The evidence I will keep

**Write or draw your reasoning and evidence.**

---

---

---

---

---

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to

keep your practice evidence together.

## Transfer practice from the original programme

A library kiosk receives 'mystery', filters a catalogue by genre and displays three titles. No speaker is attached. Draw its information path. A visitor says it read three books aloud. Which stage contradicts that claim?

**Explain the mechanism, evidence and one limit.**

---

---

---

---

---

---

---

## Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.